

Duality II: Optimality and Complementary Slackness

A feasible solution and a feasible dual bound can meet. When they do, we have a complete optimality certificate.

Our route: strong duality $\rightarrow$ a numerical certificate $\rightarrow$ complementary slackness $\rightarrow$ the simplex connection.

We continue with minimization problems; their duals maximize lower bounds.

Practice writing duals

Strong duality

Complementary slackness

Recover the dual weights

Optional proofs and extensions

How to construct the dual

- 1. Objective:** a minimization primal $\min c^\top x$ gives a maximization dual $\max b^\top p$.
- 2. One weight per row.** Row i of A is $a_i^\top$; its constraint sets the sign of p_i .

Primal constraint	Dual-variable restriction
$a_i^\top x \geq b_i$	$p_i \geq 0$
$a_i^\top x \leq b_i$	$p_i \leq 0$
$a_i^\top x = b_i$	p_i unrestricted

- 3. One dual constraint per variable.** $(A^\top p)_j$ is the weighted coefficient of x_j .

Primal-variable restriction	Dual constraint
$x_j \geq 0$	$(A^\top p)_j \leq c_j$
$x_j \leq 0$	$(A^\top p)_j \geq c_j$
x_j unrestricted	$(A^\top p)_j = c_j$

- 4. Check:** include the objective, every constraint, and every variable restriction.

Practice: write the complete dual

8–10 minutes. Work individually or with a partner. Include the objective, constraints, and variable restrictions. You do not need to solve the problems.

Problem A

$$\begin{array}{ll}\min & 2x_1 + 3x_2 \\ \text{s.t.} & x_1 + x_2 \geq 4, \quad x_1 + 2x_2 \geq 6, \quad x_1, x_2 \geq 0.\end{array}$$

Problem B

$$\begin{array}{ll}\min & x_1 + 2x_2 \\ \text{s.t.} & x_1 - x_2 \leq 3, \quad 2x_1 + x_2 \geq 2, \\ & x_1 \text{ unrestricted}, \quad x_2 \geq 0.\end{array}$$

Problem C

$$\begin{array}{ll}\min & 13x_1 + 10x_2 + 6x_3 \\ \text{s.t.} & 5x_1 + x_2 + 3x_3 = 8, \quad 3x_1 + x_2 = 3, \quad x_1, x_2, x_3 \geq 0.\end{array}$$

All three primals minimize. Use the sign guide on the previous page; answers follow.

Problem A: nonnegative variables and $\geq$ rows

Problem A

$$\begin{array}{ll}\min & 2x_1 + 3x_2 \\ \text{s.t.} & x_1 + x_2 \geq 4, \quad x_1 + 2x_2 \geq 6, \quad x_1, x_2 \geq 0.\end{array}$$

Dual

$$\begin{array}{ll}\max & 4p_1 + 6p_2 \\ \text{s.t.} & p_1 + p_2 \leq 2, \\ & p_1 + 2p_2 \leq 3, \quad p_1, p_2 \geq 0.\end{array}$$

Weight signs: both primal rows are $\geq$, so $p_1, p_2 \geq 0$.

Dual constraints: both primal variables are nonnegative, so both weighted coefficients must be $\leq$ their cost coefficients.

Problem B: mixed rows and a free variable

Problem B

$$\begin{array}{ll}\min & x_1 + 2x_2 \\ \text{s.t.} & x_1 - x_2 \leq 3, \quad 2x_1 + x_2 \geq 2, \\ & x_1 \text{ unrestricted}, \quad x_2 \geq 0.\end{array}$$

Dual

$$\begin{array}{ll}\max & 3p_1 + 2p_2 \\ \text{s.t.} & p_1 + 2p_2 = 1, \\ & -p_1 + p_2 \leq 2, \quad p_1 \leq 0, \quad p_2 \geq 0.\end{array}$$

Weight signs: row 1 is $\leq$, so $p_1 \leq 0$; row 2 is $\geq$, so $p_2 \geq 0$.

Dual constraints: unrestricted x_1 requires equality; $x_2 \geq 0$ gives a $\leq$ comparison.

Problem C: equality rows and a matching bound

Problem C

$$\begin{array}{ll}\min & 13x_1 + 10x_2 + 6x_3 \\ \text{s.t.} & 5x_1 + x_2 + 3x_3 = 8, \quad 3x_1 + x_2 = 3, \quad x_1, x_2, x_3 \geq 0.\end{array}$$

Dual

$$\begin{array}{ll}\max & 8p_1 + 3p_2 \\ \text{s.t.} & 5p_1 + 3p_2 \leq 13, \quad p_1 + p_2 \leq 10, \quad 3p_1 \leq 6, \\ & p_1, p_2 \text{ unrestricted.}\end{array}$$

Weight signs: equality rows allow unrestricted weights. **Dual constraints:** all $x_j \geq 0$, so all three comparisons are $\leq$.

The feasible points $x = (1, 0, 1)$ and $p = (2, 1)$ both have value 19. Weak duality certifies optimality; we will check the arithmetic shortly.

Next question: if a primal has a finite optimum, must a matching dual solution always exist?

Strong duality: the best bound reaches the best cost

$$\begin{aligned} (P) \quad & \min c^\top x && \text{s.t. } Ax = b, \ x \geq 0, \\ (D) \quad & \max b^\top p && \text{s.t. } A^\top p \leq c, \ p \text{ unrestricted.} \end{aligned}$$

Theorem 4.4 — assumption: the primal LP is feasible and has a finite optimal value.

Conclusion: both primal and dual attain optimal solutions, and their optimal objective values are equal.

$$c^\top x^* = b^\top p^*.$$

The same statement holds starting from a finite dual optimum. Strong duality does not require nondegeneracy.

What can happen to a primal–dual pair?

Primal minimization	Dual finite optimum	Dual unbounded above	Dual infeasible
Finite optimum	Possible: equal values	Impossible	Impossible
Unbounded below	Impossible	Impossible	Possible
Infeasible	Impossible	Possible	Possible

A finite optimum on one side gives a finite optimum on the other. If one problem is unbounded, the other must be infeasible.

An infeasible primal does not by itself prove that the dual is unbounded. See an example in which both are infeasible.

Why simplex can produce the matching bound

At a final simplex dictionary: every nonbasic reduced cost is nonnegative.

The row combinations used to form the objective row also give weights on the original equalities.

Nonnegative reduced costs say that these weights are dual feasible. At the current basic solution, the weighted right-hand sides equal the current objective.

Weak duality checks a proposed certificate; strong duality guarantees that a matching certificate exists. The full proof is in the appendix.

Check the proposed primal solution explicitly

For the primal $\min 13x_1 + 10x_2 + 6x_3$, use $x^* = (1, 0, 1)$.

$$\begin{aligned} 5(1) + 0 + 3(1) &= 8, \\ 3(1) + 0 &= 3, \\ x^* &\geq 0. \end{aligned}$$

$$c^\top x^* = 13(1) + 10(0) + 6(1) = 19.$$

This verifies primal feasibility and an upper bound of 19. Now check the lower-bound certificate.

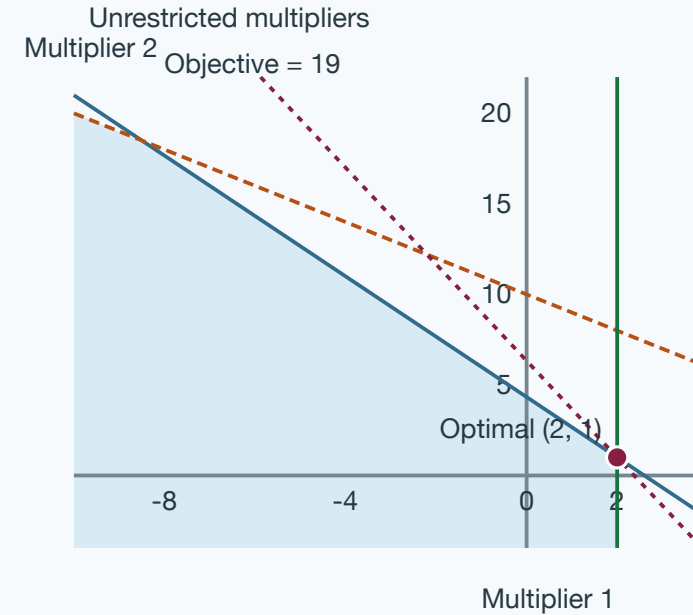
Check the proposed dual solution explicitly

Try $p^* = (2, 1)$. The dual maximizes $8p_1 + 3p_2$, with unrestricted weights.

$$\begin{aligned} 5(2) + 3(1) &= 13 \leq 13, \\ 2 + 1 &= 3 \leq 10, \\ 3(2) &= 6 \leq 6. \end{aligned}$$

$$b^\top p^* = 8(2) + 3(1) = 19.$$

The point lies in the dual feasible region. Its lower bound matches the primal cost.



Solid blue: constraint 1; dashed orange: constraint 2; green vertical: constraint 3; dotted maroon: objective 19. The feasible region continues beyond the window.

The certificate is a short arithmetic identity

For every primal feasible x , the equalities are $5x_1 + x_2 + 3x_3 = 8$ and $3x_1 + x_2 = 3$.

$$\begin{aligned} 13x_1 + 10x_2 + 6x_3 &= 2(5x_1 + x_2 + 3x_3) + (3x_1 + x_2) + 7x_2 \\ &= 19 + 7x_2 \geq 19. \end{aligned}$$

Our feasible point has $x_2 = 0$, so it attains the lower bound. Thus the primal and dual optimum is 19.

The unused amount 7 in the second dual inequality explains the extra term in this identity.

The objective gap is the sum of variable–slack products

For $Ax = b$, $x \geq 0$, let p be dual feasible and define its **dual slacks** by $r = c - A^\top p \geq 0$.

$$\begin{aligned} c^\top x - b^\top p &= c^\top x - p^\top Ax \\ &= (c - A^\top p)^\top x \\ &= \sum_j r_j x_j \geq 0. \end{aligned}$$

The gap is zero exactly when $r_j x_j = 0$ for every variable j . Each summand is nonnegative.

Next we also allow slack in the primal constraints.

With inequalities, both sides have slack

$$\begin{aligned}(P) \quad & \min c^\top x && \text{s.t. } Ax \geq b, \ x \geq 0, \\(D) \quad & \max b^\top p && \text{s.t. } A^\top p \leq c, \ p \geq 0.\end{aligned}$$

Define the primal slacks $s = Ax - b \geq 0$ and the dual slacks $r = c - A^\top p \geq 0$.

Here $p \geq 0$ because it weights $\geq$ constraints. In our equality example the weights are unrestricted and $s = 0$.

Add and subtract the same quantity to expose both gaps

For a feasible pair in the inequality form, add and subtract $p^\top Ax$:

$$\begin{aligned} c^\top x - b^\top p &= c^\top x - p^\top Ax + p^\top Ax - p^\top b \\ &= (c - A^\top p)^\top x + p^\top (Ax - b) \\ &= \sum_j r_j x_j + \sum_i p_i s_i. \end{aligned}$$

Every term is nonnegative. The total gap is zero exactly when every one of these products is zero.

Complementary slackness: two feasible solutions are optimal

Use the pair $\min\{c^\top x : Ax \geq b, x \geq 0\}$ and $\max\{b^\top p : A^\top p \leq c, p \geq 0\}$.

Theorem 4.5 — assumption: x and p are feasible for their respective problems.

Conclusion: they are both optimal if and only if

$$\begin{aligned} p_i(a_i^\top x - b_i) &= 0 && \text{for every row } i, \\ x_j(c_j - A_j^\top p) &= 0 && \text{for every variable } j. \end{aligned}$$

At optimality the total gap is zero; since every product is nonnegative, each must vanish.

Read each zero product carefully

Product	If the first factor is positive...	If the slack is positive...
$p_i s_i = 0$	$p_i > 0 \Rightarrow s_i = 0$: the primal row is tight.	$s_i > 0 \Rightarrow p_i = 0$: its weight is zero.
$x_j r_j = 0$	$x_j > 0 \Rightarrow r_j = 0$: the dual row is tight.	$r_j > 0 \Rightarrow x_j = 0$: that variable is unused.

A zero first factor does not force positive slack. Both factors may be zero.

Our equality example has only the variable–slack products

For $x^* = (1, 0, 1)$ and $p^* = (2, 1)$, the dual slacks are $r = (0, 7, 0)$.

Variable	x_j	$r_j = c_j - A_j^\top p$	$x_j r_j$
x_1	1	0	0
x_2	0	7	0
x_3	1	0	0

All primal rows are equalities: $Ax - b = 0$. Thus $p_i(a_i^\top x - b_i) = 0$ automatically, even though p_i can have either sign.

The positive variables 1 and 3 identify the dual constraints that must be tight.

Recover the weights from the positive variables

Suppose we know the optimal primal point $x^* = (1, 0, 1)$. Complementary slackness gives

$$\begin{aligned}x_1^* > 0 &\Rightarrow 5p_1 + 3p_2 = 13, \\x_3^* > 0 &\Rightarrow 3p_1 = 6.\end{aligned}$$

$$p_1 = 2, \quad p_2 = 1.$$

Finally check the remaining dual inequality: $p_1 + p_2 = 3 \leq 10$. Solving the tight equations proposes a candidate; this last feasibility check completes the certificate.

The matrix equation packages the same two equalities

The basic variables are x_1, x_3 , so their columns and costs are

$$B = \begin{pmatrix} 5 & 3 \\ 3 & 0 \end{pmatrix}, \quad c_{\mathcal{B}} = \begin{pmatrix} 13 \\ 6 \end{pmatrix}.$$

$$B^{\top} p = c_{\mathcal{B}} \iff \begin{cases} 5p_1 + 3p_2 = 13, \\ 3p_1 = 6. \end{cases}$$

This is the equation we just solved. Equivalently, $p^{\top} = c_{\mathcal{B}}^{\top} B^{-1}$. No explicit inverse is needed here.

When all basic variables are positive, their tight constraints determine these weights uniquely.

The simplex sign test is a dual-feasibility test

For basis B , compute p from $B^\top p = c_B$. Then

$$\bar{c}_j = c_j - c_B^\top B^{-1} A_j = c_j - p^\top A_j.$$

Thus $\bar{c} \geq 0$ is exactly $A^\top p \leq c$: dual feasibility.

If the current basic solution is also primal feasible, its cost equals the weighted right-hand sides. The feasible pair with equal values is an optimality certificate.

An optimal point may be degenerate. We need a basis whose reduced costs pass the sign test; an arbitrary basis representing that point need not do so.

Tight equations alone can propose the wrong weights

Return to the same primal equations. Another feasible basic solution is $x = (0, 3, 5/3)$, with cost 40.

$$5(0) + 3 + 3(5/3) = 8, \quad 3(0) + 3 = 3.$$

Its positive variables are 2 and 3. Making their dual inequalities tight gives

$$p_1 + p_2 = 10, \quad 3p_1 = 6 \quad \Rightarrow \quad p = (2, 8).$$

Does this candidate satisfy every dual inequality?

The missing dual inequality rejects that candidate

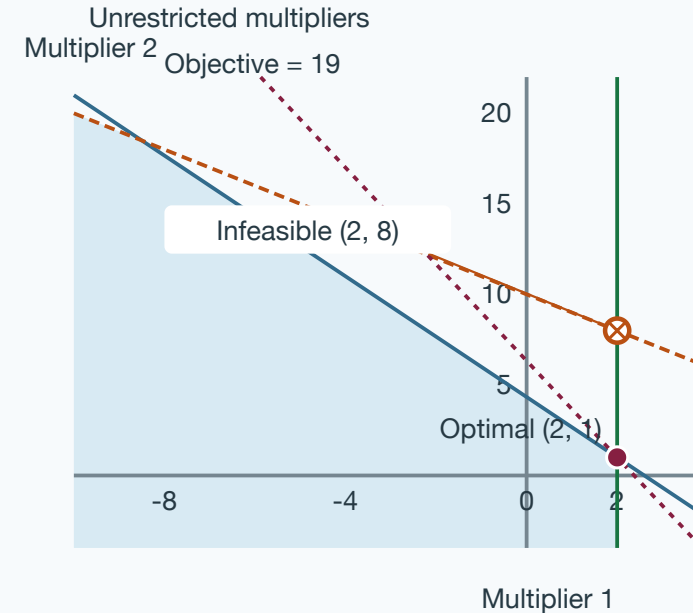
At $p = (2, 8)$, the first dual inequality fails:

$$5p_1 + 3p_2 = 34 > 13.$$

Its dual slack, and the reduced cost of x_1 , is $13 - 34 = -21$.

Although $b^\top p = 40 = c^\top x$, this is not a valid lower bound: p is infeasible.

Equal values and complementary products cannot replace the feasibility checks.



Solid blue: constraint 1; dashed orange: constraint 2; green vertical: constraint 3; dotted maroon: objective 19. The feasible region continues beyond the window.

Practice: construct and check a complete certificate

$$\begin{array}{ll}\min & 2x_1 + 3x_2 \\ \text{s.t.} & x_1 + x_2 \geq 4, \quad x_1, x_2 \geq 0.\end{array}$$

Try the primal point $x = (4, 0)$.

1. Write the one-variable dual.
2. Use the positive primal variable to find a candidate weight.
3. Check both feasibility and equality of the objective values.

Then check both kinds of complementary products.

Practice answer: one weight certifies a cost of 8

$$\max 4p \quad \text{s.t.} \quad p \leq 2, \quad p \leq 3, \quad p \geq 0.$$

Since $x_1 = 4 > 0$, its dual constraint is tight: $p = 2$. This satisfies every dual constraint.

$$2(4) + 3(0) = 8 = 4(2).$$

Both points are feasible and their values match, so both are optimal.

$$p s = 2(4 - 4) = 0, \quad x_1 r_1 = 4(2 - 2) = 0, \quad x_2 r_2 = 0(3 - 2) = 0.$$

Use three checks before claiming optimality

1. Primal feasibility. Check every original constraint and variable sign.

2. Dual feasibility. Check every weight sign and dual constraint.

3. Matching values. Compare objective values, or check all complementary products.

Bertsimas and Tsitsiklis §4.3, Theorems 4.4–4.5; Example 4.6.

Next: interpret these same weights as prices and use weighted constraints to certify infeasibility.

Optional study: complete proofs, degeneracy, and additional examples

Optional proofs and extensions

Strong duality: complete simplex proof

Example 4.5: both problems can be infeasible

Complementary slackness: necessity and sufficiency

Why degeneracy changes multiplier recovery

An inactive constraint and its zero weight

Feasible-set size versus objective boundedness

Strong duality proof: start with a terminating simplex run

$$\begin{aligned} (P) \quad & \min c^\top x && \text{s.t. } Ax = b, \ x \geq 0, \\ (D) \quad & \max b^\top p && \text{s.t. } A^\top p \leq c, \ p \text{ unrestricted.} \end{aligned}$$

First assume A has full row rank and the primal is feasible with a finite optimum.

Phase I supplies a feasible basis. Run Phase II with a valid anticycling rule. The LP is not unbounded, so the run terminates at a feasible basis whose reduced costs are nonnegative.

$$x_{\mathcal{B}} = B^{-1}b \geq 0, \quad x_{\mathcal{N}} = 0, \quad c^\top - c_{\mathcal{B}}^\top B^{-1}A \geq 0.$$

Basic values may be zero. The argument uses the final reduced-cost signs, not a nondegeneracy assumption.

Strong duality proof: the final basis gives feasible weights

Define p by $B^\top p = c_{\mathcal{B}}$, or equivalently $p^\top = c_{\mathcal{B}}^\top B^{-1}$.

$$c^\top - p^\top A = c^\top - c_{\mathcal{B}}^\top B^{-1} A \geq 0.$$

$$A^\top p \leq c.$$

These are exactly the dual constraints. Equality rows impose no sign restriction on p .

Every basic dual inequality is tight. The nonbasic inequalities hold because the final reduced costs are nonnegative.

Strong duality proof: the objective values agree

At this feasible basic solution, $x_{\mathcal{N}} = 0$ and $x_{\mathcal{B}} = B^{-1}b$. Therefore

$$\begin{aligned} b^{\top} p &= c_{\mathcal{B}}^{\top} B^{-1} b \\ &= c_{\mathcal{B}}^{\top} x_{\mathcal{B}} \\ &= c^{\top} x. \end{aligned}$$

We have primal feasibility, dual feasibility, and matching values. Weak duality proves that both solutions are optimal.

This proves strong duality for full-row-rank standard form, including degenerate optima.

Extend the proof: row signs and dependent equalities

Replace $a_i^\top x \geq b_i$ by $a_i^\top x - s_i = b_i$, $s_i \geq 0$. Its zero-cost slack column gives $-p_i \leq 0$, hence $p_i \geq 0$.

Replace $a_i^\top x \leq b_i$ by $a_i^\top x + s_i = b_i$. Its slack column gives $p_i \leq 0$. Equality rows keep p_i unrestricted.

For a feasible equality system, discard dependent rows until the remaining rows are independent. Extend any weights for the retained rows by zero on the discarded rows.

Both $A^\top p$ and $b^\top p$ are unchanged by that extension, so the same certificate works for the original system.

Extend the proof: variable signs and the objective convention

For a free variable, write $x_j = u_j - v_j$, with $u_j, v_j \geq 0$. The two dual inequalities are

$$A_j^\top p \leq c_j, \quad -A_j^\top p \leq -c_j,$$

which together say $A_j^\top p = c_j$.

For $x_j \leq 0$, substitute $x_j = -u_j$. The dual inequality becomes $A_j^\top p \geq c_j$.

These substitutions preserve feasible solutions and objective values. They recover the general dual sign rules from standard form.

For a maximization primal, negate the objective to use the minimization proof, then undo that sign change. Dualizing the dual recovers the original problem, giving the symmetric theorem.

Example 4.5: both the primal and dual can be infeasible

In this example, the primal variables and the equality weights are unrestricted.

$$\begin{array}{ll} (P) & \min x_1 + 2x_2 \quad \text{s.t. } x_1 + x_2 = 1, \quad 2x_1 + 2x_2 = 3, \\ (D) & \max p_1 + 3p_2 \quad \text{s.t. } p_1 + 2p_2 = 1, \quad p_1 + 2p_2 = 2. \end{array}$$

The primal equations disagree: twice the first left-hand side would have to equal both 2 and 3.

The dual equations also disagree: their identical left-hand sides would have to equal both 1 and 2.

Thus primal infeasibility alone does not imply dual unboundedness.

Complementary slackness proof: both directions

For feasible x, p in the inequality form, set $s = Ax - b \geq 0$, $r = c - A^\top p \geq 0$.

$$c^\top x - b^\top p = \sum_j x_j r_j + \sum_i p_i s_i.$$

Necessity. If both points are optimal, strong duality says the gap is zero. Every summand is nonnegative, so each product must be zero.

Sufficiency. If every product is zero, the gap is zero. The two feasible objective values agree; weak duality makes both points optimal.

For equality rows the primal slack is identically zero; their unrestricted multipliers cause no extra condition. The variable–dual–slack terms give the same conclusion.

A zero basic variable need not make its dual constraint tight

$$\min x_1 \quad \text{s.t. } x_1 = 0, \quad x_1 \geq 0.$$

The only basic solution is $x_1 = 0$: it is optimal and degenerate. Its dual is

$$\max 0p \quad \text{s.t. } p \leq 1, \quad p \text{ unrestricted.}$$

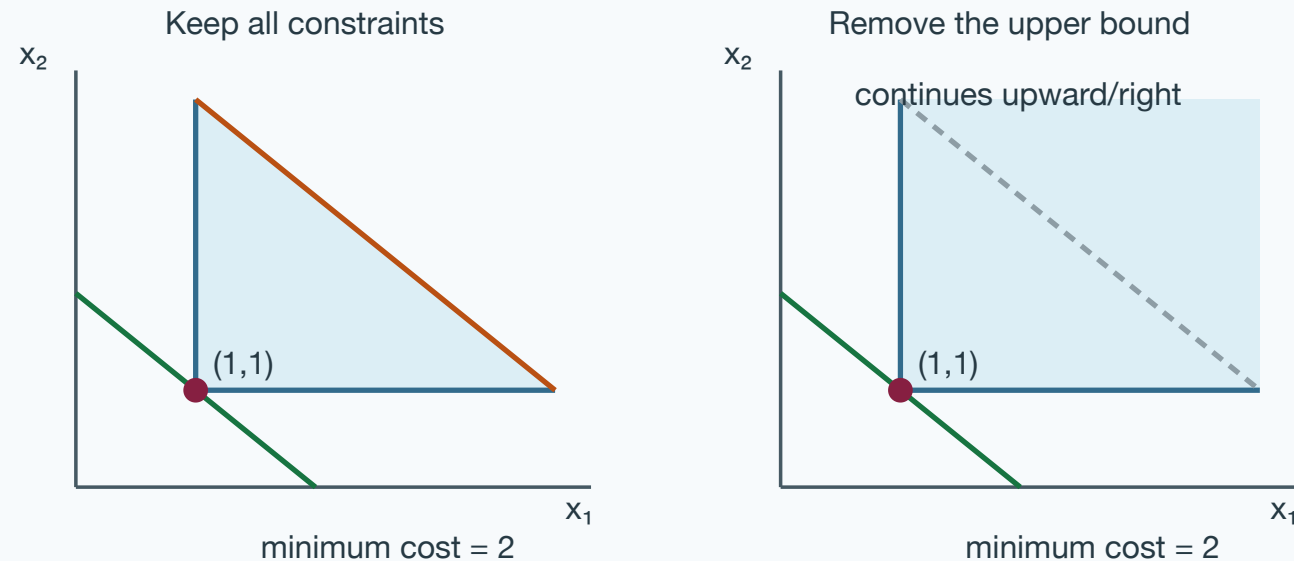
Complementary slackness says $0(1 - p) = 0$, which holds for every dual feasible p . It does not determine a unique multiplier.

The basis equation $B^\top p = c_{\mathcal{B}}$ chooses $p = 1$, one valid optimal certificate. Strong duality still holds.

In a nondegenerate optimum, every basic variable is positive, so its dual constraint must be tight; that is why the recovery argument is stronger there.

An inactive constraint can have zero weight at optimum

$$\min x_1 + x_2 \quad \text{s.t. } x_1 \geq 1, x_2 \geq 1, x_1 + x_2 \leq 5.$$



At $(1, 1)$, the upper bound has slack 3 and its optimal multiplier is zero. Removing that bound enlarges the region but leaves the minimum 2.

Why removing one inactive inequality preserves this optimum

Let x^* be optimal and suppose one inequality is strict at x^* . Remove that inequality.

If a newly allowed point y had lower cost, then $x(t) = (1 - t)x^* + ty$ would also have lower cost for every $t > 0$.

For sufficiently small $t > 0$, the removed inequality would still hold because it had positive slack at x^* . All other constraints hold along the segment by convexity.

That would give a cheaper point in the original feasible set, contradicting optimality.

This explains the zero multiplier: the inactive inequality is unnecessary for a tight lower-bound certificate.

An unbounded feasible set is not an unbounded objective

Clark's theorem: unless both members of a primal–dual LP pair are infeasible, at least one has an unbounded feasible set. See textbook Exercise 4.21.

For example, $\min\{x : x \geq 0\}$ has an unbounded feasible ray but a finite optimum 0.

“Unbounded feasible set” concerns where feasible points can go. “Unbounded minimization problem” means feasible objective values decrease without limit. These statements are different.

This result is additional theory; the optimality certificate still uses the same three checks.